

Theorem 18.5.2. For random variables R_1, R_2 and constants $a_1, a_2 \in \mathbb{R}$,

$$\text{Ex}[a_1 R_1 + a_2 R_2] = a_1 \text{Ex}[R_1] + a_2 \text{Ex}[R_2].$$

In other words, expectation is a linear function. A routine induction extends the result to more than two variables:

Corollary 18.5.3 (Linearity of Expectation). For any random variables $R_1, \dots, R_k$ and constants $a_1, \dots, a_k \in \mathbb{R}$,

$$\text{Ex} \left[\sum_{i=1}^k a_i R_i \right] = \sum_{i=1}^k a_i \text{Ex}[R_i].$$

The great thing about linearity of expectation is that *no independence is required*. This is really useful, because dealing with independence is a pain, and we often need to work with random variables that are not known to be independent.

As an example, let's compute the expected value of the sum of two fair dice.

18.5.1 Expected Value of Two Dice

What is the expected value of the sum of two fair dice?

Let the random variable R_1 be the number on the first die, and let R_2 be the number on the second die. We observed earlier that the expected value of one die is 3.5. We can find the expected value of the sum using linearity of expectation:

$$\text{Ex}[R_1 + R_2] = \text{Ex}[R_1] + \text{Ex}[R_2] = 3.5 + 3.5 = 7.$$

Assuming that the dice were independent, we could use a tree diagram to prove that this expected sum is 7, but this would be a bother since there are 36 cases. And without assuming independence, it's not apparent how to apply the tree diagram approach at all. But notice that we did *not* have to assume that the two dice were independent. The expected sum of two dice is 7—even if they are controlled to act together in some way—as long as each individual controlled die remains fair.

18.5.2 Sums of Indicator Random Variables

Linearity of expectation is especially useful when you have a sum of indicator random variables. As an example, suppose there is a dinner party where n men check their hats. The hats are mixed up during dinner, so that afterward each man receives a random hat. In particular, each man gets his own hat with probability $1/n$. What is the expected number of men who get their own hat?

Letting G be the number of men that get their own hat, we want to find the expectation of G . But all we know about G is that the probability that a man gets